

# **CARISURG Triage Guardian**

## **Week 9 Deployment System Requirements**

Project: Clinical AI Risk Identification System for Urgent Triage

Student: Shari Oliver

Programme: CariSurg MedTech Pathways

Design Focus: Human-Computer Interaction (HCI), with Human-Robot Interaction (HRI) concept exploration

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This document defines how Triage Guardian is intended to be deployed, the observable requirements governing its human-machine interface, the data it consumes and emits, and the selected build path for the Week 9 prototype. The HCI dashboard is the preferred initial implementation; an HRI pre-triage kiosk is retained as an alternative design concept for comparison.

## 1. System Purpose and Deployment Vision

Triage Guardian is an AI-assisted clinical decision support system designed for emergency department (ED) triage. Its purpose is to help triage nurses identify patients who may require rapid assessment by combining patient demographics, presenting complaints, physiological observations, and relevant clinical history into a risk-informed recommendation.

The system is not intended to replace clinical judgement or autonomously assign final triage categories. The clinician remains the decision-maker. The AI acts as a second layer of support by highlighting potentially high-risk patients, showing the factors that contributed to the recommendation, and making changes in patient status easier to recognise.

### Preferred deployment: HCI nurse-facing dashboard

- The primary deployment is a secure web-based dashboard accessible from an ED triage work station or approved clinical tablet.
- The dashboard presents a live triage queue, patient status, vital-sign information, AI-generated risk alerts, and an explanation of the factors contributing to each alert.
- The system is positioned within the existing triage workflow: data are entered or retrieved, the model processes the available information, the nurse reviews the output, and the nurse accepts, modifies, or overrides the recommendation.
- The design prioritises minimal disruption to the current workflow and maintains clear clinician control.

### Alternative deployment concept: HRI pre-triage kiosk

- The HRI concept is a physical pre-triage kiosk positioned near ED intake.
- Patients could enter basic demographic and symptom information using a touchscreen, while connected devices could capture selected physiological measurements where appropriate.
- The kiosk would transmit preliminary information and risk flags to the triage nurse rather than independently making a final clinical decision.
- This concept requires additional hardware, accessibility controls, infection-control procedures, sensor reliability safeguards, and physical-space planning; therefore, it is not the preferred Week 9 build path.

## 2. Users and Human-Machine Interaction

**Primary HCI user:** ED triage nurse. The nurse reviews the queue, opens patient records, interprets AI alerts, compares the recommendation with clinical findings, and determines the final triage action.

**Secondary HCI users:** physicians and other authorised clinical staff who may review patient status, risk flags, or the reasoning associated with an escalation.

**HRI users:** patients interact directly with the kiosk during preliminary intake, while nurses receive and validate the information generated by that interaction.

### 3. Explicit System Requirements

The following requirements are written as observable behaviours so that they can be evaluated during prototyping or later implementation.

| ID     | Area                  | Observable requirement                                                                                                                                                                              |
|--------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HCI-01 | Queue visibility      | The interface must display all active triage patients in a single queue view with patient identifier, waiting time, current status, and alert level visible without opening each record.            |
| HCI-02 | Alert differentiation | The interface must visually distinguish high-risk, moderate-risk, and lower-risk states using both colour and text/icon labels so that risk is not communicated by colour alone.                    |
| HCI-03 | Clinical control      | Every AI recommendation must be presented as decision support. The nurse must be able to accept, modify, or override the recommendation before a final triage action is recorded.                   |
| HCI-04 | Explainability        | When an AI alert is selected, the interface must display the principal patient factors contributing to the recommendation, such as abnormal vital signs, presenting complaint, or relevant history. |
| HCI-05 | Timeliness            | After the system receives a complete new set of required triage inputs, the interface should display or refresh the model output within 2 seconds under normal operating conditions.                |
| HCI-06 | Critical change       | If new physiological data cause a patient to cross a configured high-risk threshold, the queue must update the patient's alert state and visibly indicate that the status has changed.              |
| HCI-07 | Acknowledgement       | A clinician must be able to acknowledge a high-risk alert, and the system must retain the alert state until it has been reviewed or superseded by newer information.                                |
| HCI-08 | Auditability          | The system must record the model recommendation, timestamp, clinician response, and any override so that the sequence of decisions can be reviewed.                                                 |
| HCI-09 | Legibility            | Core triage information and alert labels must remain readable at the intended workstation display size without requiring the user to zoom the page.                                                 |
| HCI-10 | Accessibility         | Critical information must not rely solely on colour; text labels, symbols, or patterns must provide an additional cue.                                                                              |

|        |                      |                                                                                                                                                                                                          |
|--------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HCI-11 | Data minimisation    | The interface must display only information required for the triage task and restrict access to authorised users.                                                                                        |
| HCI-12 | Failure state        | If the model output cannot be generated because data are missing or the service is unavailable, the interface must state that AI support is unavailable rather than presenting a misleading risk result. |
| HRI-01 | Patient confirmation | Before kiosk-entered information is transmitted, the patient must be given an opportunity to review or confirm key entered details.                                                                      |
| HRI-02 | Clinical hand-off    | The kiosk must send collected information to the nurse-facing workflow and must not independently finalise the patient's clinical triage category.                                                       |
| HRI-03 | Sensor failure       | If a connected sensor fails or produces an invalid reading, the kiosk must identify the affected measurement as unavailable and continue safely without fabricating a value.                             |
| HRI-04 | Physical safety      | The kiosk must use a stable, accessible form factor with no moving component that can obstruct, pinch, or strike the patient during normal use.                                                          |
| HRI-05 | Privacy              | Patient information displayed on the kiosk must be limited to the active intake interaction and cleared from the public-facing screen when the session ends.                                             |

## 4. Model Inputs

The prototype assumes a combination of manual clinical entry, available electronic health record (EHR) information, and physiological measurements. Not every source must be available in the Week 9 mock-up; the list defines the intended integration vision.

### Manual clinical entry

- Patient identifier and basic demographics
- Presenting complaint
- Pain score
- Arrival mode
- Nurse observations and clinical notes

### Physiological/device data

- Heart rate
- Blood pressure
- Respiratory rate

- Oxygen saturation (SpO2)
- Temperature

### **EHR-derived context**

- Relevant previous ED visits
- Known medical history
- Current or past medications where available
- Relevant allergies or other clinical flags

## **5. Model Outputs and Human Response**

- Risk/priority indication: a clearly labelled risk state or prioritisation recommendation shown in the triage queue.
- Alert state: a visible warning when the available data indicate a potentially high-risk patient or meaningful deterioration.
- Supporting explanation: a short display of the factors that most influenced the model recommendation.
- Status update: refreshed output when new patient information or vital signs are entered.
- Uncertainty/failure message: an explicit indication when required data are incomplete or the AI service cannot provide a valid result.

### **Human action after model output**

The triage nurse reviews the AI output alongside the patient's clinical presentation. The nurse may prioritise immediate assessment, continue the current triage pathway, collect additional information, or override the AI recommendation. The interface therefore supports clinical reasoning rather than substituting for it. Where an override occurs, the action should be retained in the audit trail.

## **6. End-to-End HCI Workflow**

1. Patient presents to the emergency department and enters the triage workflow.
2. The nurse enters the presenting complaint and observations; available EHR and device data are retrieved or added.
3. The system validates whether the minimum required inputs are present.

4. The AI model processes the available information and produces a risk-informed recommendation.
5. The dashboard updates the patient's queue status and presents the recommendation and supporting factors.
6. The nurse reviews the output against the clinical assessment.
7. The nurse accepts, modifies, or overrides the recommendation and proceeds with the appropriate triage action.
8. New information or vital signs trigger reassessment and an updated display where applicable.
9. Relevant AI outputs and clinician actions are logged for traceability.

## 7. Selected Build Path and Justification

Selected path: Web-based HCI interface. The Week 9 prototype will prioritise a nurse-facing web dashboard rather than a mobile-only interface or physical HRI prototype.

- A web interface aligns closely with the ED workstation environment and the current dashboard mock-up.
- It can present a queue of multiple patients simultaneously, which is central to triage prioritisation.
- It supports rapid iteration of alert states, explanations, patient cards, and workflow controls without requiring specialised hardware.
- It preserves clinician oversight and can be designed to integrate with EHR and device data sources in a future implementation.
- Compared with the HRI kiosk, it introduces fewer physical-safety, infection-control, accessibility, maintenance, and sensor-integration requirements for an initial prototype.

## 8. HCI-HRI Deployment Comparison

| Design factor       | HCI dashboard                            | HRI kiosk                                               |
|---------------------|------------------------------------------|---------------------------------------------------------|
| Primary interaction | Clinician reviews and acts on AI support | Patient enters preliminary data; nurse reviews hand-off |
| Form factor         | Web application on workstation/tablet    | Physical kiosk with touchscreen and possible sensors    |
| Clinical authority  | Triage nurse                             | Triage nurse                                            |
| Workflow disruption | Low to moderate                          | Moderate to high                                        |

|                     |                                            |                                                                    |
|---------------------|--------------------------------------------|--------------------------------------------------------------------|
| Hardware dependency | Existing computing hardware                | Dedicated kiosk and connected devices                              |
| Key safety risks    | Alarm fatigue, legibility, automation bias | Physical accessibility, sensor failure, privacy, infection control |
| Week 9 build status | Preferred prototype                        | Conceptual alternative/mock-up                                     |

## 9. Safety, Privacy and Operational Constraints

### Alarm fatigue

Repeated low-value alerts can reduce attention to important warnings. The design should reserve the most prominent alert state for clinically meaningful risk and avoid unnecessary repeated notifications.

### Automation bias

Clinicians may over-trust an AI recommendation. The interface must frame outputs as recommendations, show supporting factors, and retain an obvious clinician override pathway.

### Display legibility

Triage decisions are made in a time-pressured environment. Risk state, waiting time, and critical observations must be visually prominent and understandable at a glance.

### Data privacy

Patient information must be available only to authorised users, with minimal necessary information exposed on shared screens and appropriate session controls.

### Data quality

Missing, delayed, or incorrect measurements can distort model output. The system must identify unavailable or invalid data rather than silently treating them as normal.

### System downtime

Clinical triage must remain possible if the AI service is unavailable. The dashboard must communicate failure clearly and allow the conventional clinical workflow to continue.

## HRI-specific constraints

A physical kiosk would additionally require wheelchair-accessible positioning, screen and control reachability, cleaning between users, management of noisy environments, sensor validation, and a safe fallback when hardware is unavailable.

## 10. Prototype Acceptance Criteria

- A reviewer can identify the active triage queue and distinguish patient alert states immediately.
- At least one high-risk patient state is visibly represented in the HCI mock-up.
- A reviewer can see what information the AI receives and what output is returned to the clinician.
- The design clearly communicates that the nurse retains final decision authority.
- The prototype includes an interaction path from patient selection to review of risk information or alert reasoning.
- The HRI mock-up clearly depicts a physical pre-triage form factor and its hand-off to the clinical workflow.
- The two concepts are visibly differentiated rather than being the same interface placed on different devices.

## 11. Implementation Boundary for Week 9

Week 9 focuses on design and prototyping rather than production deployment. The deliverable therefore demonstrates the intended workflow, human-machine interaction, inputs, outputs, alert behaviour, and deployment logic. It does not claim live EHR integration, validated medical-device connectivity, or autonomous clinical operation. These would require further technical development, clinical validation, governance, security review, and usability testing.

## 12. Conclusion

Triage Guardian is designed primarily as a web-based HCI clinical decision support tool for emergency triage. The system receives clinical and physiological information, generates a risk-informed recommendation and alert state, explains the basis of the output, and returns control to the triage nurse for the final action. An HRI pre-triage kiosk has also been explored to satisfy the physical-interaction design requirement, but the HCI dashboard is selected as the preferred initial build path because it better fits the current clinical workflow, requires less specialised infrastructure, and preserves direct clinician oversight. The requirements in this document provide specific, observable behaviours against which the Week 9 prototype and future iterations can be evaluated.